

Kevin Davidson

kevin@kevindavidson.work | [LinkedIn](#) | [GitHub](#) | [Website](#) | [Hugging Face](#)

EDUCATION

University of California, San Diego B.S., Math - Computer Science	Jun. 2028
Santa Monica College A.S.-T, Mathematics	Jun. 2026
GPA: 3.7/4.0 Honors Program Member IEEE Computer Society Member	

TECHNICAL SKILLS

Languages: C++17, Python (AI/ML), JavaScript (React), Bash, MATLAB
Developer Tools: YOLOv8, Hailo-8L NPU, Linux (Debian/Ubuntu), Edge Computing, PyTorch (Model Training)
Software and Frameworks: React.js, Git/GitHub Actions, Vercel
Web & Design: Figma, Framer, Adobe Creative Suite (Illustrator, Photoshop), CAD

PROFESSIONAL EXPERIENCE

SIQ (Fintech AI Startup)	May 2026 - Present
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Machine Learning Engineer Intern

- Developing data pipelines and RAG infrastructure for an AI-powered financial research platform.
- Architected SEC EDGAR ingestion systems with semantic chunking for data preprocessing, streamlining institutional investment analysis workflows.

MathWorks	Jul. 2026 - Aug. 2026
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Computer Vision Developer Intern

- Engineered a hybrid visual inspection pipeline in MATLAB for automated manufacturing, combining classical computer vision (morphology, thresholding) with deep learning.
- Fine-tuned a ResNet-50 CNN via transfer learning to perform binary defect classification on pharmaceutical pills.
- Developed evaluation scripts to validate system robustness against factory floor variations (signal noise, Gaussian blur), achieving 90.3% accuracy on the MVTec AD dataset.

Santa Monica College Santa Monica, CA	Sep. 2025 - Dec. 2025
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Supplemental Instruction Leader/ Teaching Assistant

- Directed peer-to-peer instruction for a 120-student Data Structures (C++) course**, focusing on complex memory management, pointers, and OOP paradigms.
- Debugged 50+ student C++ programs weekly** using VS Code to resolve memory leaks and segmentation faults.
- Integrated **CI/CD workflows** into the curriculum, teaching **GitHub Actions** for automated testing.

RESEARCH & PROJECTS

TALOS: Open-Source Bicycle Safety System

- Achieved 26x inference speedup (3 to 80+ FPS)** by migrating the vision pipeline to a **Hailo-8L NPU**, enabling real-time blind-spot monitoring on a Raspberry Pi 5.
- Increased motorcycle class recall by 41%** by fine-tuning YOLOv8 on a custom dataset with class oversampling and published versioned models to Hugging Face.
- Built a threat-isolation algorithm** based on **Tau Theory (Time-to-Collision)** to filter background noise and calculate time-to-collision, **reducing false-positive alerts on stationary vehicles** for high-speed transit safety.

Apollo: Full-Stack Campus Resource Application | React.js, Python

- Streamlined campus navigation and study-room discovery by architecting a **live interactive SVG map** in React.js that visualizes real-time room occupancy for **250+ students monthly** at apollo.kevindavidson.work.
- Engineered a **Python ETL pipeline** to scrape and sanitize unstructured Oracle APEX schedules, converting **1700+ records** into optimized JSON for sub-second client-side rendering through **GitHub Actions**.

LEADERSHIP EXPERIENCE

HackSMC Santa Monica, CA	Jan. 2026 - Apr. 2026
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Outreach and Sponsorship Lead

- Directed external communications and strategic outreach for a regional hackathon, **leading a team of 5** to secure **\$16k** in corporate sponsorships, university grants, and prizes by pitching a tiered sponsorship system to companies.

AWARDS & HONORS

- 1st Place, SMC STEM Poster Symposium 2026
- 2nd Place, CCSC Southwest 2026
- Helset Scholar Finalist, **Stanford** Bay Honors Symposium 2026
- Honorable Mention, IEEE SusTech 2026
- Finalist, SMC Global Citizenship Symposium 2026